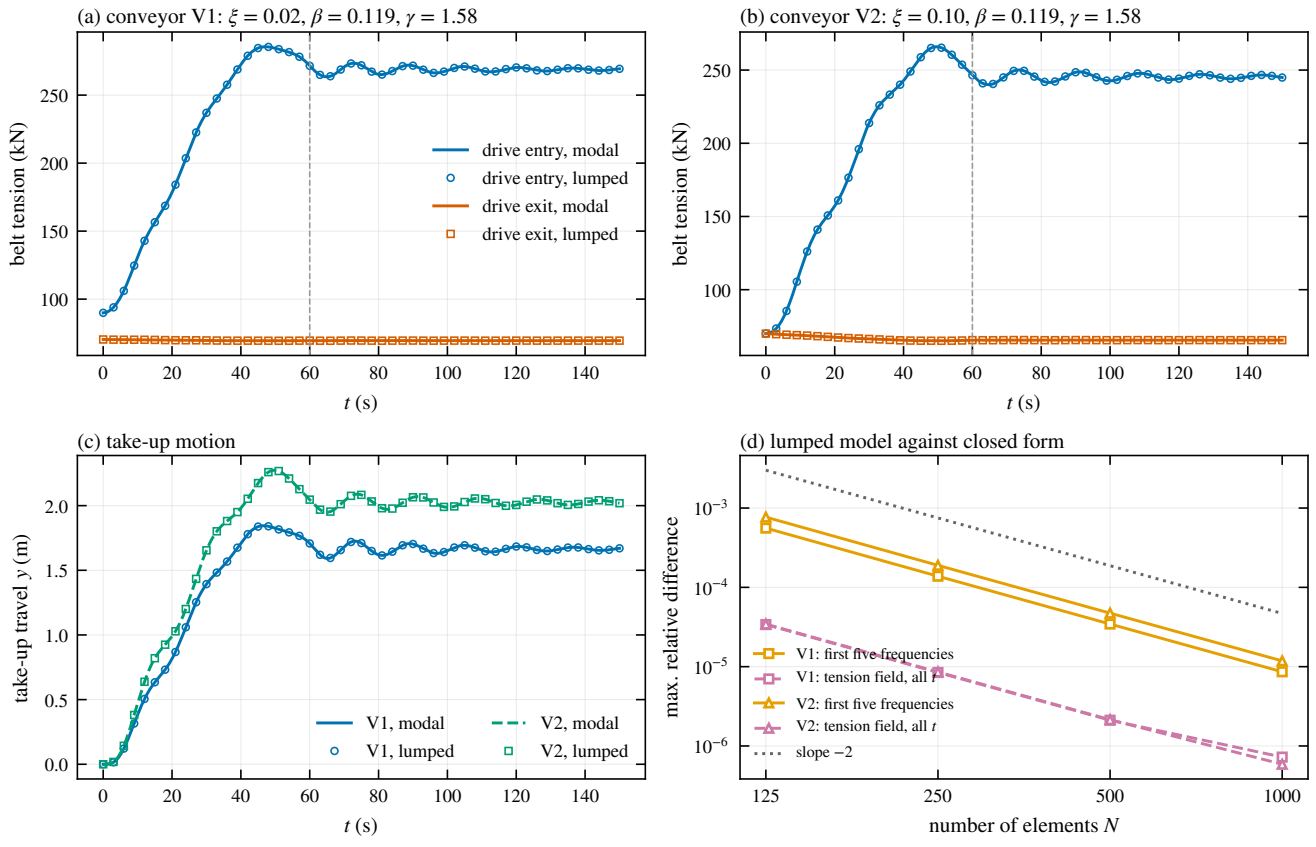
(a) conveyor V1: $\xi = 0.02$, $\beta = 0.119$, $\gamma = 1.58$
belt tension (kN)
t (s)
drive entry, modal
drive entry, lumped
drive exit, modal
drive exit, lumped
(b) conveyor V2: $\xi = 0.10$, $\beta = 0.119$, $\gamma = 1.58$
belt tension (kN)
t (s)
(c) take-up motion
take-up travel y (m)
t (s)
V1, modal
V1, lumped
V2, modal
V2, lumped
(d) lumped model against closed form
max. relative difference
number of elements N
V1: first five frequencies
V1: tension field, all t
V2: first five frequencies
V2: tension field, all t
slope −2